Geophysics: Final Project Description

For this study, you will aim to:

1. interpret the subsurface layers in light of the geologic history of the area (seismic data component), and
2. perform studies to address the problem outlined in your assigned scenario (gravity and MT component).

You will use the instruments at your disposal – a seismic unit, a gravimeter, and a magnetometer – to systematically explore your assigned area.

Assignment Overview

You will receive four different data sets from the survey efforts. You will begin with analyzing the seismic reflection and refraction data, followed by gravity and magnetic data sets of the same area. Treat the audience of your report and presentation as members of the U.S. geological survey, geotechnical engineering, or a broad academic audience.

The final project consists of three components: a presentation to check your progress, your code from analyzing your data and plotting your results, and a 12 page report. More details on each component are below.

Presentation on preliminary findings (midpoint check-in)

1. 6 minute presentation, 2 minutes for questions, for a total presentation time of 8 minutes per student. (We don't want you to go over time, but coming in a few minutes under time is okay *as long as you convey all of the necessary information*.)
2. Presentation should cover your problem statement, relevant geologic information, and a mostly complete analysis of your **seismic data. This must include your seismic cross section from SeismicCrossSection.ipynb.**
3. You may include a preliminary analysis of the gravity and magnetic data, but it is not required for your grade.
4. Include citations in your presentation if you reference any work done by someone else (classmates or journal articles, etc.).
5. Structure should follow a similar format to the report, i.e.:
    a. Introduction with problem statement,
    b. Background information,
    c. Methods (seismic methods),
    d. Present your preliminary findings for the seismic data (with figures), and then
    e. Future work on the project (namely the gravity and magnetic surveys you will be analyzing after the presentation)
    f. Brief summary of your whole presentation
6. **In order to receive credit for your presentation, you <u>MUST ask at least one question or provide feedback</u> to one of your peers during the presentation Q&A.**

Computational workflow as a Python notebook

1. One or two notebooks that demonstrate the calculations and the figures. Providing the script of your work is a central element to research reproducibility and is recommended (or imposed) by scientific journals.
2. Provide a notebook that is well-organized and commented on. Follow the structure of the report under the sections Methods and Results, and use the notebook to explain your analysis. Upload the notebook itself, rendered with all the figures, on Canvas.
3. Save the figures from the notebook in PNG and add them to your report.
4. NB: Remember that you need to import all important python packages used in the lab (pyrocko, harmonica, verde, matplotlib, numpy, pandas, …). **Do not use any packages that were not previously used in the course.**

Written report

1. Introduction to the problem
    a. Problem Statement and why geophysical surveys will help the analysis. You may cite scientific journals or USGS reports to support the problem statement.
    b. What kind of geophysical survey was performed, and what is the data availability (brief statement)? Describe the arrays of sensors and the sources used.
    c. A brief overview of this report.
    d. < 0.75 page
2. Background
    a. Area geology and context
    b. History
    c. Cite other studies
3. Methods
    a. Seismic reflection
    b. Seismic refraction
    c. Gravity
    d. Magnetics
    e. For each method, write a concise summary on how these methods work and what particular aspect of the data analysis can help constrain it. Include equations relevant to your methods.
    f. Each method should include 2-3 sentences about strengths and weaknesses of the methods/data in imaging the subsurface.
4. Results
    a. Seismic reflection
    b. Seismic refraction
    c. Gravity
    d. Magnetics
    e. Each survey yields an individual model/image.
    f. Each result should have a short text describing the figure. This section is **not** an interpretation of the results.

    g. Complete version of cross section (x,z) from your results, using a plotting function (the plotting function will be provided to you in CompleteCrossSection.ipynb).
5. Synthesis - Discussion
    a. Your interpretation of the seismic data and more shallow subsurface (great part of the report to tie-in the geology again).
    b. Your interpretation of the gravity and magnetic data.
    c. Your interpretation of your multi-disciplinary results to characterize the magmatic system. Discuss what is the most likely **single model** that could be derived from **all images/models** (use the cross section to assist with this).
    d. Discuss the potential sources of errors or limitations in the survey and what could be done next time to provide a better model.
    e. This is a place to showcase sophisticated “critical thinking”. For instance, pose and answer the question “how are these models (seismic,gravity,magnetic) consistent with each other?”
6. Summary and conclusions

Figures

Should be clearly labeled, low margins, font size of 10-12 pts in the caption and the figure labeling itself. Each axis should have a label. Any plot with more than one curve or dataset should include a legend. **You must number the figures, caption them, and reference them *at least once* in the main text.**

References

(See the Scientific Writing - Citations page for the complete information on this.) References should be listed at the end of the document, following APA/American Geophysical Union reference style and must include a DOI. You can (and probably should) use a reference manager like **Zotero** when writing your report (Zotero will generate your references for you in the style you indicate, based on what you have saved to any given folder).

Organization

Use numbered sections and subsections (e.g., 1. Introduction, 2. Background, 3.1 Seismic Reflection Methods, etc …). **If you use any equations, please number and reference them.**

Scientific Writing and Formatting

Use **Overleaf** (or just LaTeX) with the provided template for the course and upload a PDF of your completed project to Canvas. Use 12-point font for body text, with 1” margins and single line spacing, single column (one column per page). Your report should be about 12 pages long, including figures. You will be provided with an Overleaf/LaTeX template for the final report.

Rubrics

Written report: 50 points (50% of grade for Final Project)

Item	Grade %	Total
Written Mechanics Grammar, spelling, referencing (in the correct reference style), etc.	**20%**	
Clarity and Flow of Argument Is your text organized? Does each paragraph feed off of earlier paragraphs and facilitate later ones? Is some information “extra” – i.e. do you have a paragraph that doesn’t contribute to the point of what you’re writing? etc.	**20%**	
Plausibility and Strength of Support Is your argument a sound one – i.e., is it supported by the data and figures you have available in a clear, direct fashion? Have you made a clear distinction between observations and interpretations? Do you convey clearly the things you know for sure versus those you must infer, etc.?	**25%**	
Sophistication of Argument A simple argument supported by simple evidence can be sound, but we are also looking for indications that you have taken things a step or two beyond this, intellectually. **Consider the geologic setting.** You can say, “this layer is composed of loose sediment because the seismic velocity is 2000 m/s,” or you can say, “This layer is composed of loose sediment because the seismic velocity is 2000 m/s, and glaciers deposited a lot of debris in the area many years ago.” The second is a much stronger argument (and is the difference between an “A” and a “B”).	**15%**	
Adherence to Guidelines You were asked to write 12 pages, but you wrote 20– yikes! Did you include all the required sections in your report? Did you include the required figures?	**15%**	
Effectiveness of Revisions How thorough a job have you done when revising your text in response to comments and your new data? Did you incorporate the specific comments provided to you from your presentation to improve on the final report?	**5%**	
Total is presented as a percentage of 50 points		**100%**

Notebook: 30 points (30% of grade for Final Project)

Item	Grade %	Total
Notebook Organization Import modules on the top cell in the appropriate sections, make a cell per figure etc, and **remove unused cells.**	**20%**	
Reproducibility Is your code well-commented? Can your sibling or another peer in STEM re-run the notebook and understand the steps (even without understanding geophysics in general)? Remember that the person grading your code can't read your mind.	**35%**	
Correctness Did you follow the guidance from the lab and produce a correct analysis? Even if the final image is not unique, your notebook(s) should demonstrate the correct workflow, following the exercises we did in the lab. You should be generating the figures required.	**35%**	
Adherence to Guidelines You were asked to turn in one notebook for seismic data analysis and one notebook for gravity and magnetics. Additionally, you were told to copy-paste the plotting functions into your notebooks. Follow the guidelines for the notebook to get full credit.	**10%**	
`Total is presented as a percentage of 30 points`		**100%**

Presentation: 20 points (20% of grade for Final Project)

Item	Grade %	Total
Effectiveness of Presentation Was your presentation compelling to your audience? Did you convey all of the necessary information? Did the information you shared make sense in the context of your presentation?	**40%**	
Ability to Answer Questions Were you able to respond adequately to questions during the Q&A part of your presentation?	**20%**	
Clarity of Slides Were your slides clear to read? Did your slide titles make sense? Were your figures clearly labeled and explained? Were the slides too crowded (not a good thing)?	**20%**	
Adherence to Guidelines Did you include the presentation components you were asked to? Did you include the relevant plots and explain your methods? Follow the guidelines for the presentation to get full credit.	**10%**	
Stage Presence and Eloquence Did you speak clearly? Did you talk in a way that convinced your audience that you are confident in the material you are presenting?	**10%**	
`Total is presented as a percentage of 20 points`		**100%**

Please note that in order to receive credit for this component of the final project, **you must ask a question or make a comment on someone else's presentation**. Not doing so will result in a grade of zero for the presentation component of the final project.